Hamdan Ashfaq

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EDUCATION

University of California, Irvine & San Diego State University (Joint Doctoral Program) Aug 2026 – Present
Ph.D., Computational Science · Expected 2031 *Irvine & San Diego, CA*

- *Graduate coursework (Fall 2026):* Applied Mathematics for Scientists; Mathematical Modeling; Modern Physics; Quantum Physics. *Spring 2027 (planned):* Scientific Computing; Advanced Computational PDEs; Condensed Matter Physics; Machine Learning Operations.

Denison University Jan 2023 – May 2026
B.A. Computer Science & B.S. Data Analytics, Philosophy concentration *Granville, OH*

- **Cumulative GPA 3.51 / 4.0**
- Departmental honor cords in both Computer Science and Data Analytics; Denison Phi Society Inductee, top 7% of Class of 2026.

TECHNICAL SKILLS

Programming: Python, C/C++, R, SQL; Git, Linux, Jupyter, RStudio.
Machine Learning: PyTorch, TensorFlow, Keras, scikit-learn, TIMM; deep learning, self-supervised learning, CNNs, vision transformers, unsupervised clustering, dimensionality reduction.
Atomistic Modeling: DFT (GPAW, PBE), machine-learned interatomic potentials (MLIPs), equivariant graph neural networks (MACE), ASE, molecular dynamics.
Quantitative Methods: Time-series analysis, statistical and predictive modeling, mixed-integer linear programming, optimization (Gurobi), Monte Carlo and scenario analysis, numerical simulation.
Data & Computing: NumPy, Pandas, Matplotlib, OpenCV; HPC, CUDA, parallelized pipelines; PowerBI, Tableau.

RESEARCH EXPERIENCE

San Diego State University, Dogan Group Aug 2026 – Present
Graduate Researcher (advisor: Prof. Mehmet Dogan) *San Diego, CA*

- Developing machine-learned interatomic potentials and running the DFT calculations used to train them.

The Ohio State University, Center for Electron Microscopy & Analysis (CEMAS) May 2024 – Aug 2026
Visiting Fellow, Materials Science and Engineering (May 2026 – Aug 2026; advisor: Prof. Jinwoo Hwang) *Columbus, OH*

- Simulated 4D-STEM data of dry and wet amorphous SiO2 (92K and 120K atom models from Sandia National Laboratories) with Kirkland multislice (TEMSIM), giving labeled diffraction data that experiments cannot provide.
- Built the simulation pipeline on the Ohio Supercomputer Center (SLURM job arrays, resumable runs) and computed fluctuation electron microscopy variance curves V(k) to compare with experiment.

Machine Learning Research Intern (Aug 2025 – Dec 2025) *Columbus, OH*

- Wrote time-series analysis code for an ongoing Hwang lab project that tracks how the microstructure of amorphous semiconductors changes with experimental conditions and thermal processing in 4D-STEM data.
- Built sequential ML models and autocorrelation analyses to find temporal patterns in structural transformations.

Research Intern, Intel-funded OSU SURE-5 Program (May 2024 – May 2025) *Columbus, OH*

- Led the self-supervised and unsupervised analysis (K-means clustering, t-SNE) of 65,536 unlabeled 4D-STEM nanodiffraction patterns (multiple GB); the results fed into a Microscopy & Microanalysis 2024 poster.
- Developed supervised ML models of atomic structure in amorphous semiconductors, streamlined 4D-STEM data processing, and used unsupervised methods to find the most common structural patterns.

Purdue University, SURF Research Program May 2025 – Aug 2025
Engineering Research Intern (advisor: Prof. Nikhilesh Chawla) *West Lafayette, IN*

- Built a CNN with a ResNet-50 backbone and a U-Net++ style decoder that classifies fishbone, lamellar, fine and coarse microstructures in SEM images of Sn-58Bi solder with 90.14% accuracy.
- Cut microstructure analysis from hours of manual expert review to about 30 minutes of automated classification.
- Wrote the SEM preprocessing (ROI cropping, CLAHE contrast enhancement, intensity normalization, geometric augmentation) and parallelized the TensorFlow data pipeline, raising image throughput 30%.

SELECTED PROJECTS

Fine-Tuning a Foundation Interatomic Potential for Monolayer MoS2 · GitHub Oct 2026

- Ran 157 PBE calculations in GPAW and fine-tuned MACE-MP-0 on them. Force error on a held-out 600 K MD run dropped from 244 to 19 meV/Å, and fine-tuning beat a from-scratch baseline at every training set size (3 seeds).
- Showed that fine-tuning softens the elastic response and that multi-head replay fine-tuning trades force accuracy for strain accuracy. The model ran 12 ps of stable MD on a 108-atom cell.
- Found that no model trained on pristine cells handles sulfur vacancies; adding 26 vacancy structures to training cut the vacancy force error from 301 to 39 meV/Å.

Finding Order in Disorder: Fold Symmetry in 4D-STEM Patterns (Senior Capstone) · GitHub — Aug 2025 – May 2026

- Built *foldsym*, a Python pipeline that detects diffraction speckles and classifies local 2- to 6-fold symmetry in nanodiffraction patterns of CuZr metallic glass, packaged with a CLI, a synthetic data generator, 29 unit tests and CI. **Awarded Best Senior Capstone Project, Department of Data Analytics.**
- Tested it on synthetic patterns with known labels: the rule-based classifier fails beyond a 9° rotation for 6-fold and 13.5° for 4-fold symmetry, which is why the next step is a self-supervised (SimSiam) model.

AEP Energy, Client Project (Denison Data Analytics Practicum) · GitHub — Aug 2025 – Dec 2025

- Measured how accurate PJM's hourly load forecasts (up to 7 days ahead) are for AEP Energy, by forecast horizon, time of week and weather, in R and Python (MAPE, paired t-tests, chi-square tests, NOAA weather correlations).
- Found that forecasts issued 42 to 48 hours ahead had the lowest error, that PJM overpredicted demand by about 930 MW at the 0 hour horizon, and that errors peak on weekday evenings; recommended the 42 to 48 hour window for customer peak alerts in a final report to the client.

Power-Grid Unit Commitment and Dispatch Optimization — Fall 2025

- Formulated and solved a mixed-integer linear program that schedules a thermal and renewable generation fleet over 72 hours on real PJM load and renewable-availability data, with startup costs, minimum stable output, renewable availability limits and a 15% reserve margin.
- Found a base-case optimal cost of $27.96M. Under high-demand and low-renewable scenarios the fleet fell short of its reserve margin in 53 of 72 hours, with a peak deficit near 233 MW.

Volunteer Food Delivery Platform for a Regional Food Pantry Network · GitHub — Aug 2025 – Dec 2025

- Built a production React Native and Expo app with Supabase auth and a real-time backend for a nonprofit food pantry network. Owned the login and role-routing architecture, SMS delivery notifications and live GPS location sharing.

PUBLICATIONS & PRESENTATIONS

Journal Articles, Under Review

- Islam, M., Wang, M., Fantin, A., Yoo, G.H., Kim, J.Y., **Ashfaq, H.**, Park, E.S., Maass, R., Wang, Y., Fan, Y., Chung, H.S., Lee, S.C., Hwang, J. "Linking Topological Medium-Range Order and Density-Wave Coherence to Metallic Glass Ductility." *Acta Materialia* (under review, 2026). Contribution: Swin Transformer V2 classifier for weak rotational symmetries in amorphous nanodiffraction data.

Conference Presentations & Posters

- Patnaik, S., Ganju, E., Torbati Sarraf, H., **Ashfaq, H.**, Chawla, N. "Harnessing the Power of AI in 3D X-ray Microtomography." Poster, Purdue AI Research Showcase, West Lafayette, IN, 2026.
- **Ashfaq, H.**, Wu, J., Chawla, N. "Imaging and Machine Learning of Defects in Semiconductors." Oral presentation, competitively selected, Purdue SURF Research Symposium, 2025.
- Islam, M., Kim, J.Y., Yoo, G.H., Im, S., Calderon Ortiz, G., **Ashfaq, H.**, Park, E.S., Hwang, J. "In-Depth Analysis of Structural Heterogeneity in High Entropy Bulk Metallic Glasses Using 4D-STEM." Poster, competitively selected, Microscopy & Microanalysis 2024 Conference, Cleveland, Ohio.

PROFESSIONAL EXPERIENCE

Invicta Counseling — May 2023 – Aug 2023
Automation Engineering Intern (Remote) — *Islamabad, Pakistan*

- Built an RStudio and SQL tool that cut manual data processing 40%, automated inventory management with Python and API integration (70% fewer errors), and ran C/C++ system diagnostics that reduced downtime 45%.

Denison University, Information Technology Services — May 2023 – Aug 2023
Assistant Manager, Networks and Operations — *Granville, OH*

- On a three-person team, upgraded 500 of the campus's 1,700+ Aruba routers to 503H and 515 models and maintained 180+ data closets, keeping 99.9% uptime and halving network downtime.

AWARDS & HONORS

Best Senior Capstone Project, Department of Data Analytics · Departmental Honor Cords, Computer Science and Data Analytics · Denison Phi Society Inductee, top 7% of Class of 2026 · Purdue SURF Research Fellowship · Intel-funded OSU SURE-5 Research Fellowship.

RELEVANT COURSEWORK

Machine Learning & Computing: Machine Learning; Deep Learning Theory; Algorithm Design & Analysis; Data Structures; Mathematical Foundations of CS; Computer Organization; Software Engineering.
Statistics & Optimization: Applied Statistics & Probability; Time Series & Analysis; Advanced Predictive Methods; Advanced Prescriptive Methods; Linear Programming; Mathematical Modeling; Linear Algebra & Differential Equations; Calculus II, III.

LANGUAGES

Urdu (native) · English (fluent) · Hindi (fluent) · Punjabi (fluent) · Arabic and Persian (read and write) · French (elementary).